

EE P 564: TinyML

Lecture 8: American Sign Language Recognition

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**ELECTRICAL & COMPUTER
ENGINEERING**

UNIVERSITY of WASHINGTON



Topics Covered

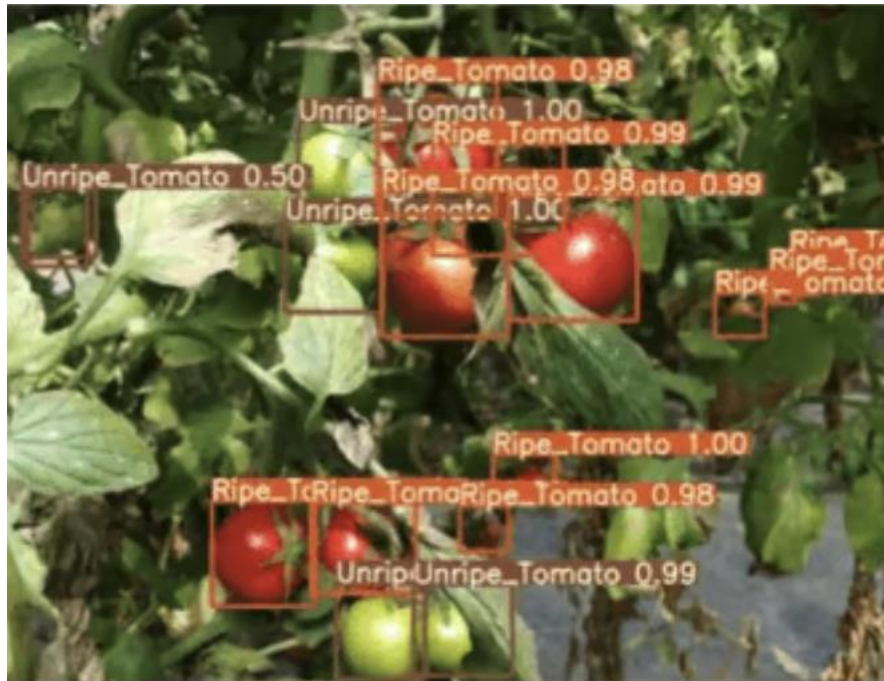
- Edge Impulse Tricks for Your Projects
- Object Detection using YOLO (for your projects; not included in lab)
- Background on ASL and ASL Interpretation
- Gesture Based Dataset and Model
- TinyML Framework for ASL Interpretation
- Lab 8: Model Training, Evaluation Metrics, and Deployment



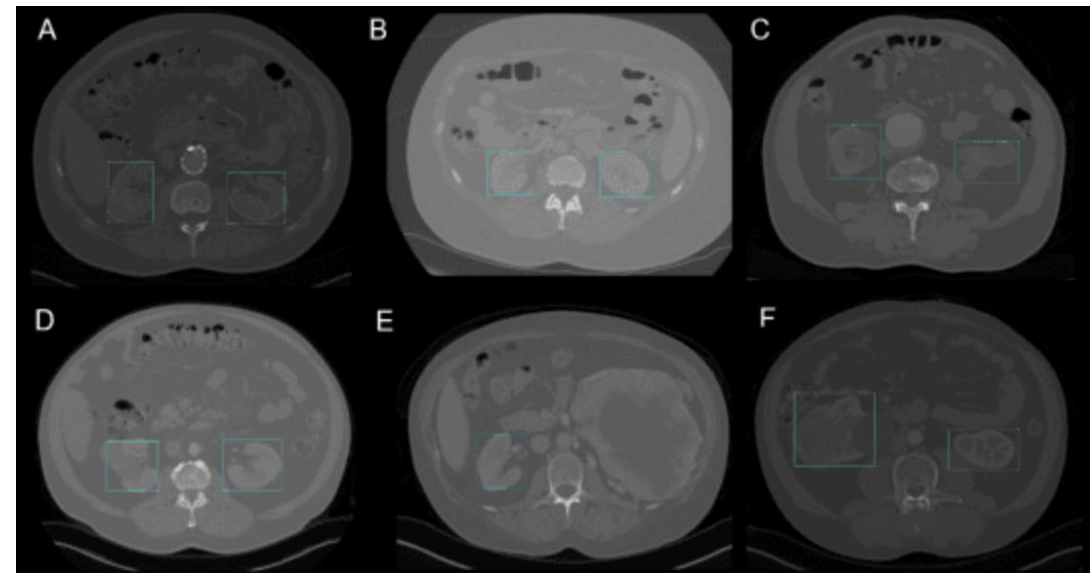
❖ Relevant readings from the Textbook [Introduction to TinyML by Rohit Sharma] is Chapter 3

Object Detection

- Detect **what** and **where** objects are in an image
- Output = **Bounding Boxes** + **Class Labels**
- **Common applications:** autonomous vehicles, security, retail, medical imaging



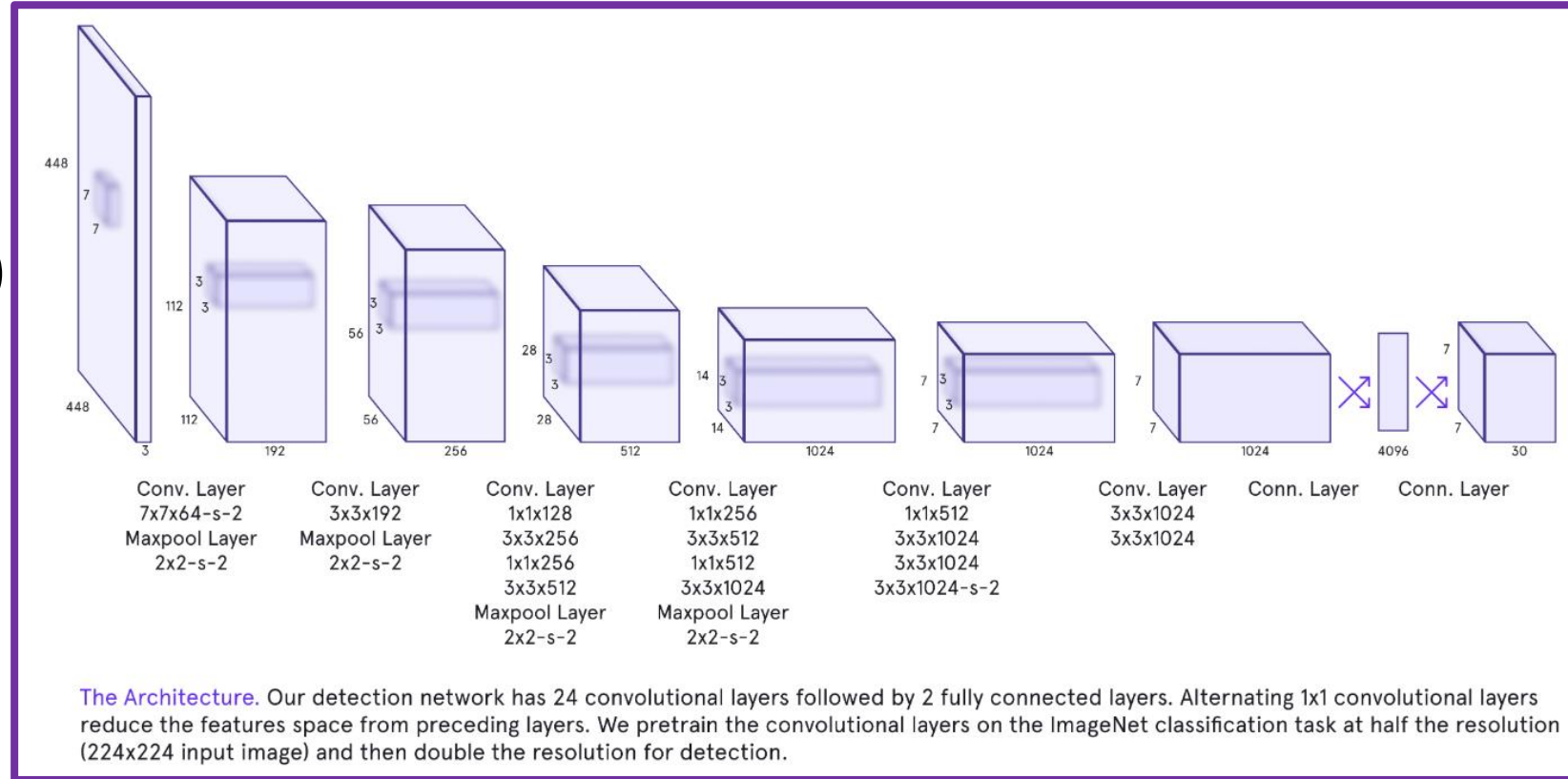
Tomato detection based on modified YOLOv3 framework



Kidney Recognition in CT using YOLOv3

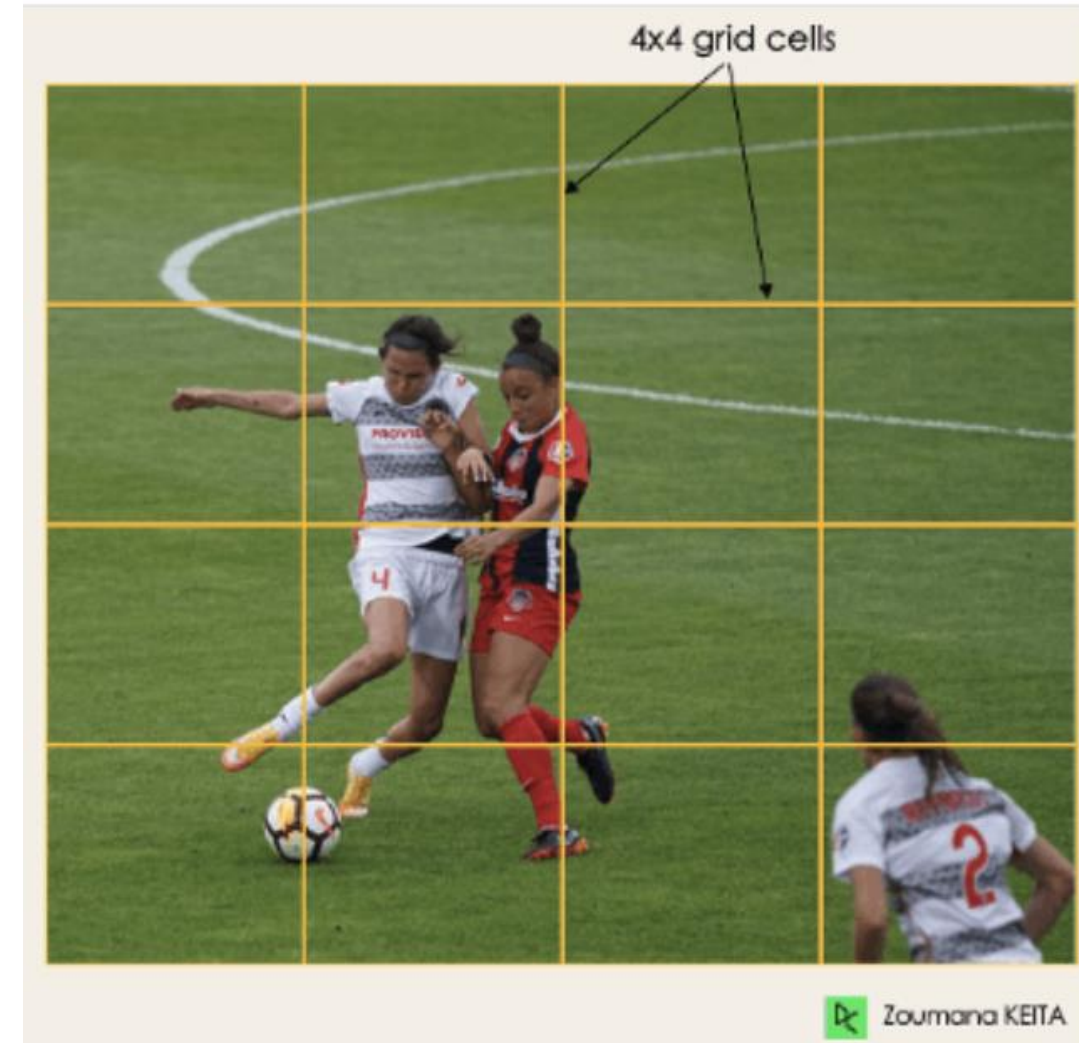
Object Detection using YOLO

- YOLO = **You Only Look Once**
- YOLO is an **object detection model** that performs:
 - Object classification**
(e.g., "cat", "car", "person")
 - Bounding box regression**
(i.e., predicting the box coordinates: center x & y coordinates, width, height)
- Single-shot end-to-end framework **based on fully convolutional neural network**



Grid Cells of Input Image

- YOLO divides image into $S \times S$ grid
- Each cell predicts:
 - **Bounding box:** (x_center, y_center, width, height)
 - **Confidence score** (objectness)
 - **Class probabilities**



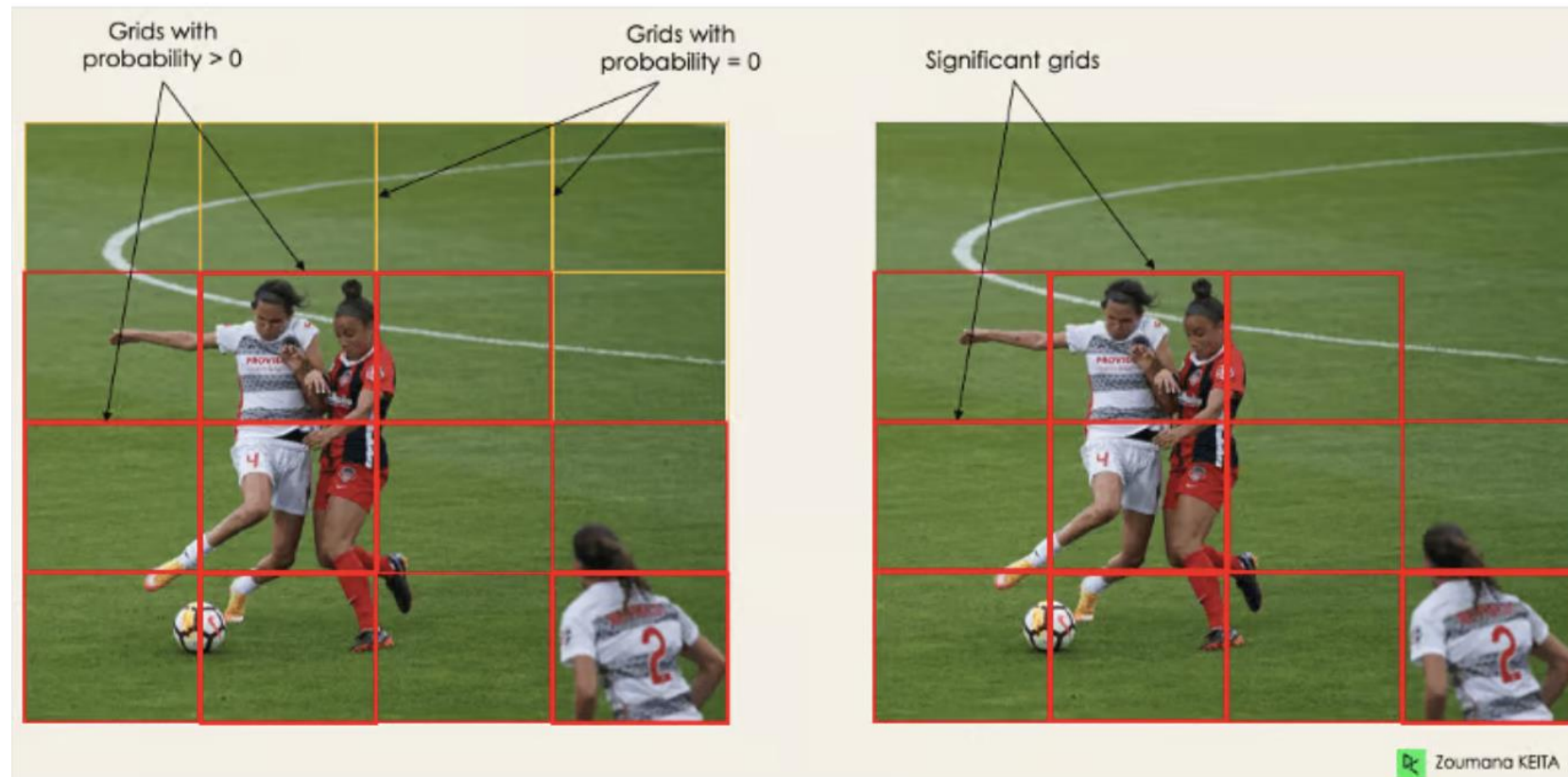
Bounding Box Regression: YOLO Output Tensor

For a single bounding box per cell and **two classes**, the output at each grid cell is:
[pc, bx, by, bw, bh, c1, c2]

[pc, bx, by, bw, bh, c1, c2]

pc (objectness score):

- **Probability that an object's center lies in this grid cell**
- Ranges from 0 to 1
- If no object is present, pc = 0
- If an object is present, $pc \approx IoU$



Bounding Box Regression: YOLO Output Tensor

bx, by: [pc, bx, by, bw, bh, c1, c2]

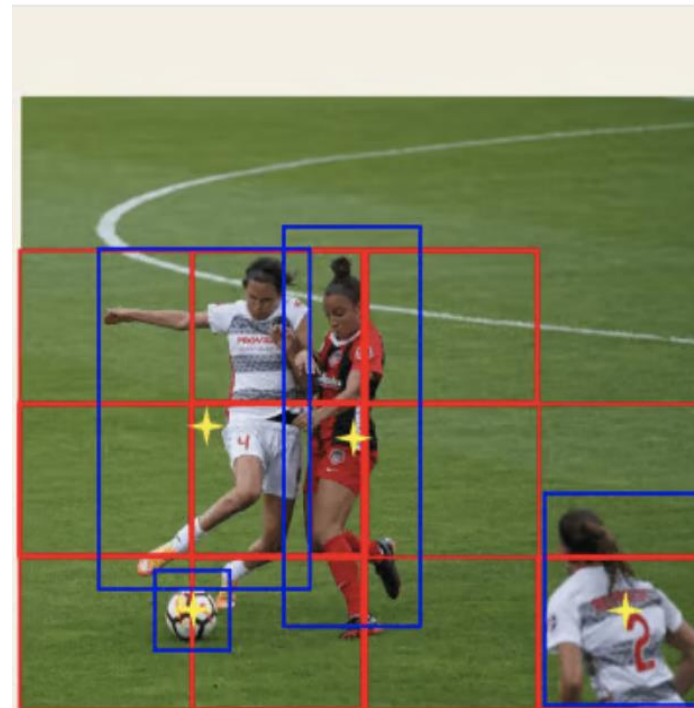
- Coordinates of the **center of the predicted bounding box, relative to the grid cell**

- Ranges from 0 to 1 within the cell

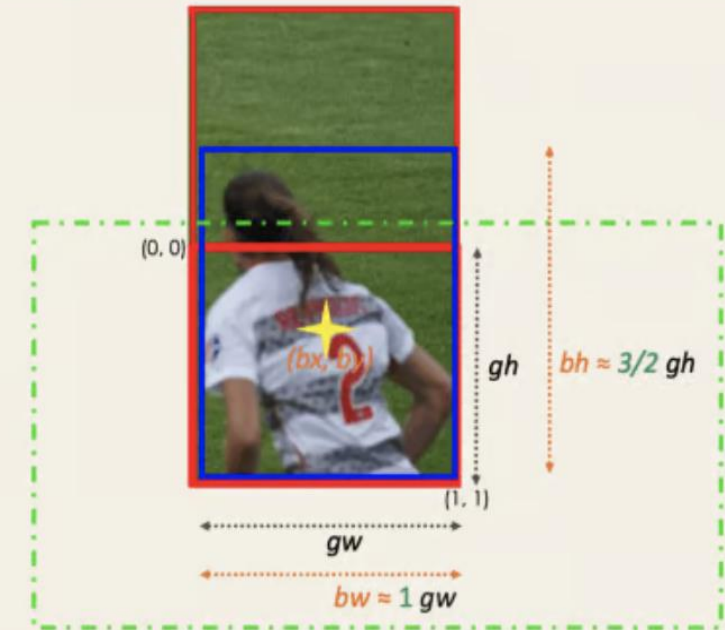
bw, bh: [pc, bx, by, bw, bh, c1, c2]

- Width and height of the predicted bounding box**

- Usually normalized by the image width/height, not just the cell size



✦ Bounding box centers



From the previous info we can have for e.g.
 $Y = [1, bx, by, 3/2, 1, c1, c2]$

- First 1 means 100% of object presence

- gh, gw: height & width of the grid
- $0 \leq bx \leq 1$
- $0 \leq by \leq 1$
- bh and bw can be more than 1

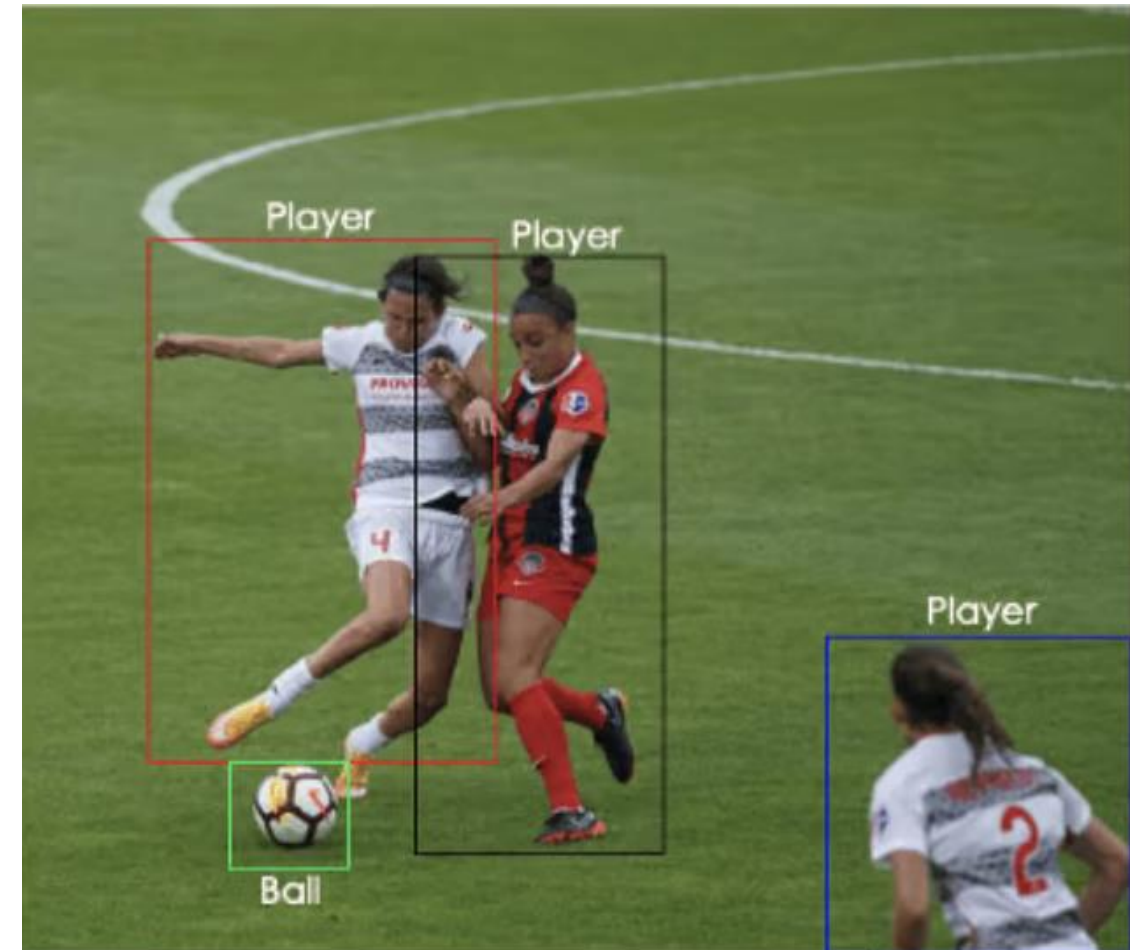
✦ Zaumana KEITA



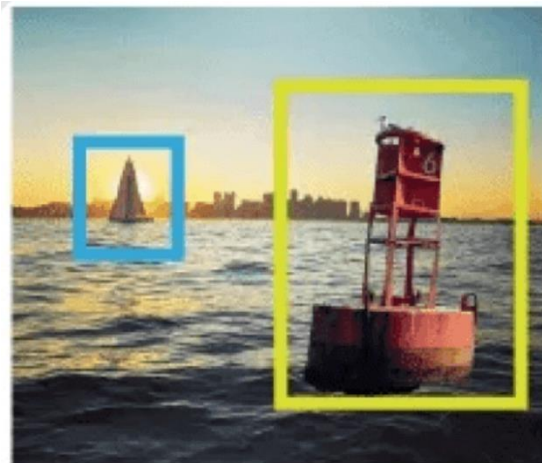
Bounding Box Regression: YOLO Output Tensor

c1,c2: [pc, bx, by, bw, bh, c1, c2]

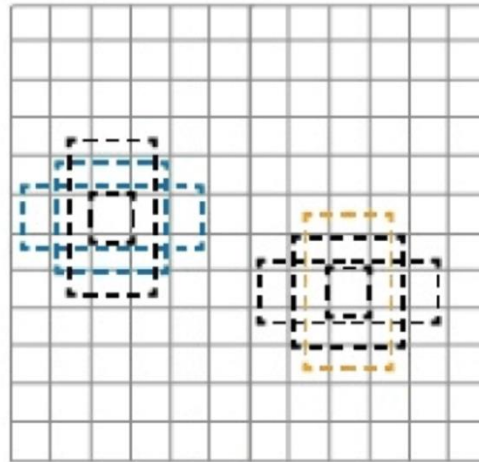
- Class probabilities: $\text{Pr}(\text{Class } c_{ic_ici} \mid \text{Object})$
- Softmax or sigmoid can be applied depending on version
- In your example: classes are **Player** and **Ball**



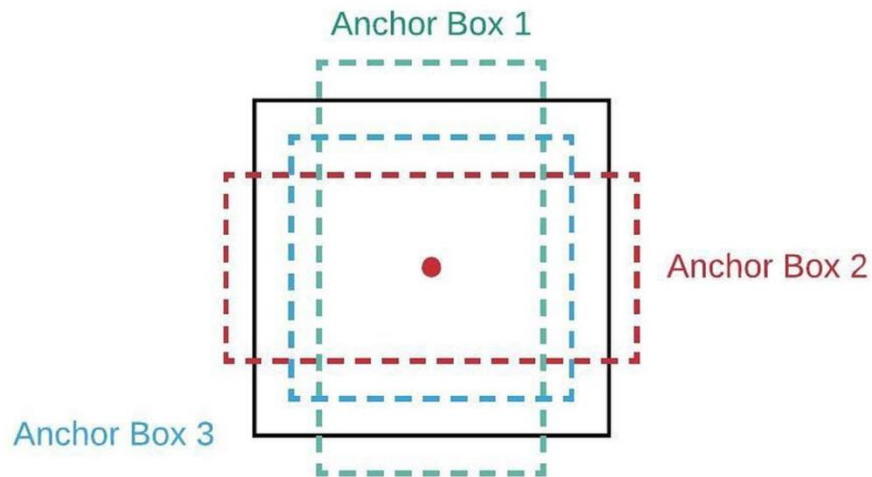
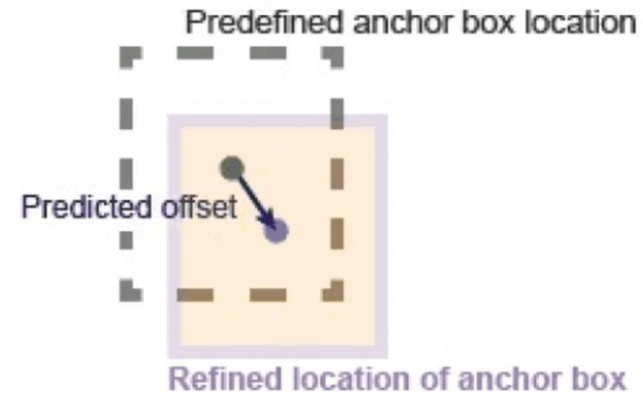
Bounding Box Regression: YOLO Output Tensor with Anchor Boxes



Ground truth image and bounding boxes



Anchor boxes at each predefined location in each feature map



Grid Cell	Grid Cell	Grid Cell
Grid Cell	Grid Cell	Grid Cell
Grid Cell	Grid Cell	Grid Cell

🧠 Why Anchor Boxes?

Different objects come in different shapes:

- A **person** tends to be tall and narrow.
- A **car** tends to be wide and short.
- A **ball** is more square.

Bounding Box Regression: YOLO Output Tensor with Anchor Boxes

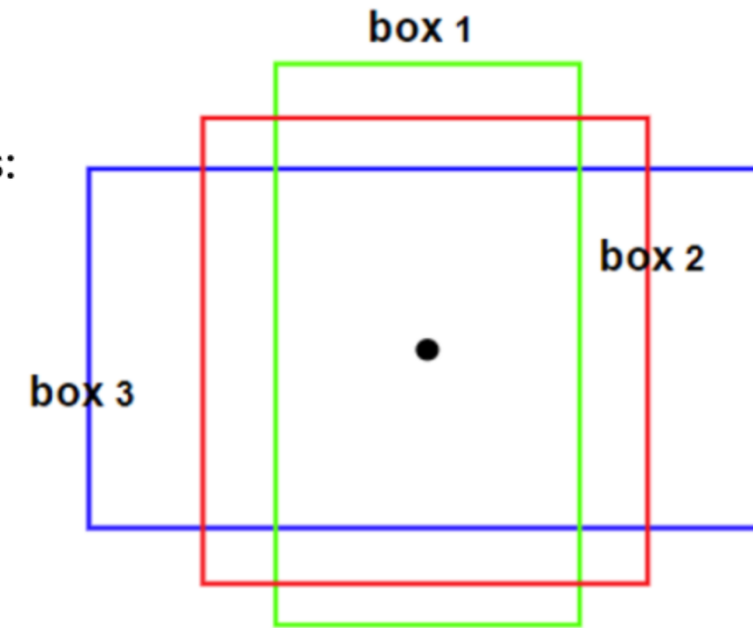
In modern YOLO versions:

- Multiple **anchor boxes** per grid cell (e.g., 3) are used → output per cell becomes:

$$[p_c, b_x, b_y, b_w, b_h, c_1, \dots, c_K] \times B$$

- So, tensor shape becomes:

$$S \times S \times B \times (5 + K)$$



Example:

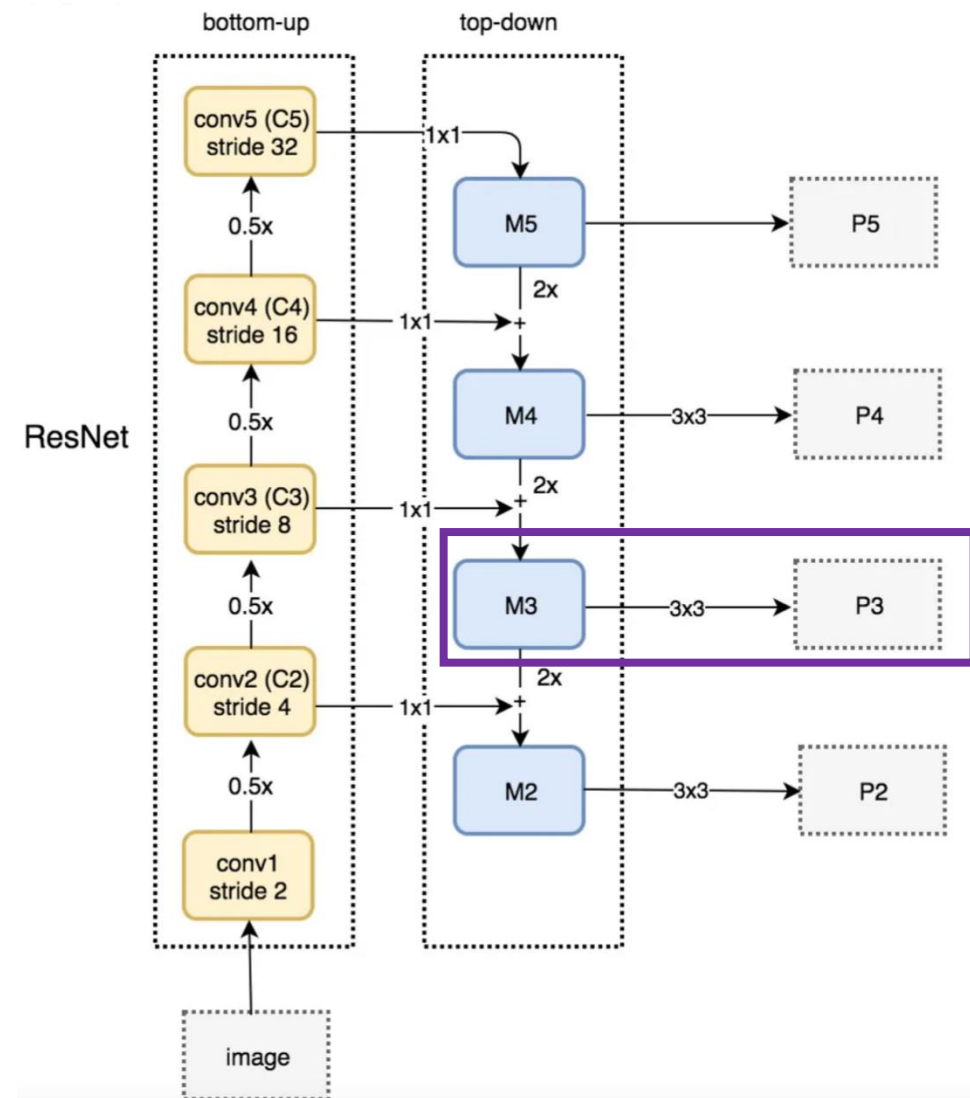
Anchor ID	Width	Height	Aspect Ratio
A1	2.0	1.0	Landscape
A2	1.0	2.0	Portrait
A3	1.0	1.0	Square

Bounding Box Regression: YOLO Output Activations

Output	Activation Used	Value Range
<code>bx</code> , <code>by</code>	Sigmoid	[0, 1] (within grid cell)
<code>bw</code> , <code>bh</code>	Exponential of linear	> 0 (scale adjusted by anchor)
<code>p_c</code> (objectness)	Sigmoid	[0, 1]
Class scores	Sigmoid / Softmax	[0, 1]

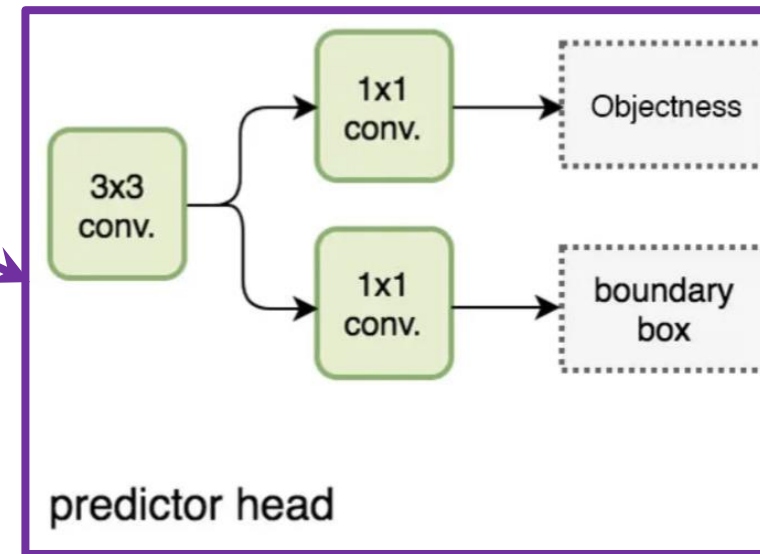


Bounding Box Regression using Feature Pyramid Networks for YOLO v3 and Up



Why FPN is important:

- Small objects occupy **fewer pixels**, so we need **higher-resolution feature maps** (like 52×52).
- Large objects can be detected at **lower resolution** (like 13×13).
- FPN enables **multi-scale predictions**, increasing robustness across object sizes.



Bounding Box Regression: Intersection Over Union (IoU)

- Measures **overlap between predicted and ground truth** box
- **IoU** = Area of Overlap / Area of Union
- Used in two key parts of YOLO:
 - **Training**: IoU is used to assign predicted boxes to the most appropriate ground-truth boxes
Some versions (e.g., YOLOv5+) use IoU-based loss (e.g., **GIoU**, **CIoU**) for better box regression
 - **Post-processing**: Used in **Non-Maximum Suppression (NMS)** to remove overlapping predictions



Training YOLO: What You Need to Provide

For each image:

- The image file (e.g., `image001.jpg`)
- A corresponding annotation file (e.g., `image001.txt`) containing:

txt

Copy

Edit

```
<class_id> <x_center> <y_center> <width> <height>
```

All values are **normalized** (between 0 and 1), relative to image dimensions.

Example line in YOLO format:

txt

Copy

Edit

```
0 0.521 0.433 0.223 0.341
```

This means: class 0, box centered at (52.1%, 43.3%) of the image, with 22.3% width and 34.1% height.



YOLO for Object Detection: Bounding Box References

Tools to draw bounding boxes:

You can manually label images using tools like:

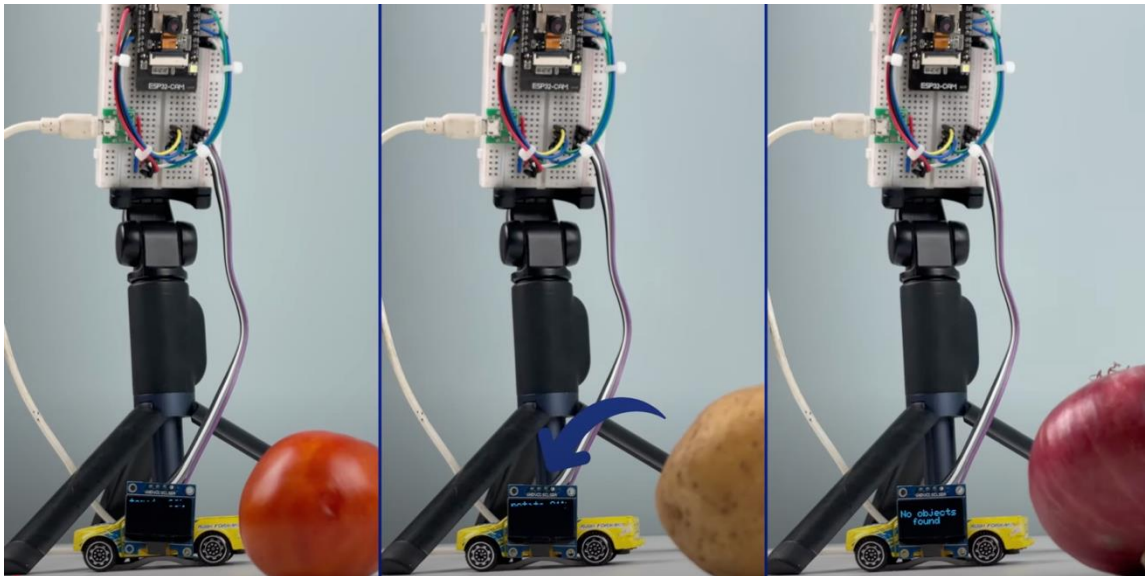
- **LabelImg** (popular and easy): <https://github.com/tzutalin/labelImg>
- **Roboflow**: web-based, easier with large datasets
- **CVAT** or **Makesense.ai**

These tools export in YOLO format directly.



YOLO for Object Detection: Edge Impulse References

- <https://www.youtube.com/watch?v=bZIKVaD3dRk>
- <https://github.com/edgeimpulse/ml-block-yolov5>
- <https://peter-ing.medium.com/beginners-guide-to-object-detection-with-edge-impulse-c8ea95f844a0>



Beginners guide to Object Detection with Edge Impulse



Peter Ing

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22 min read · Apr 17, 2021



Product ▾

Solutions ▾

Resources ▾



edgeimpulse / ml-block-yolov5

Public



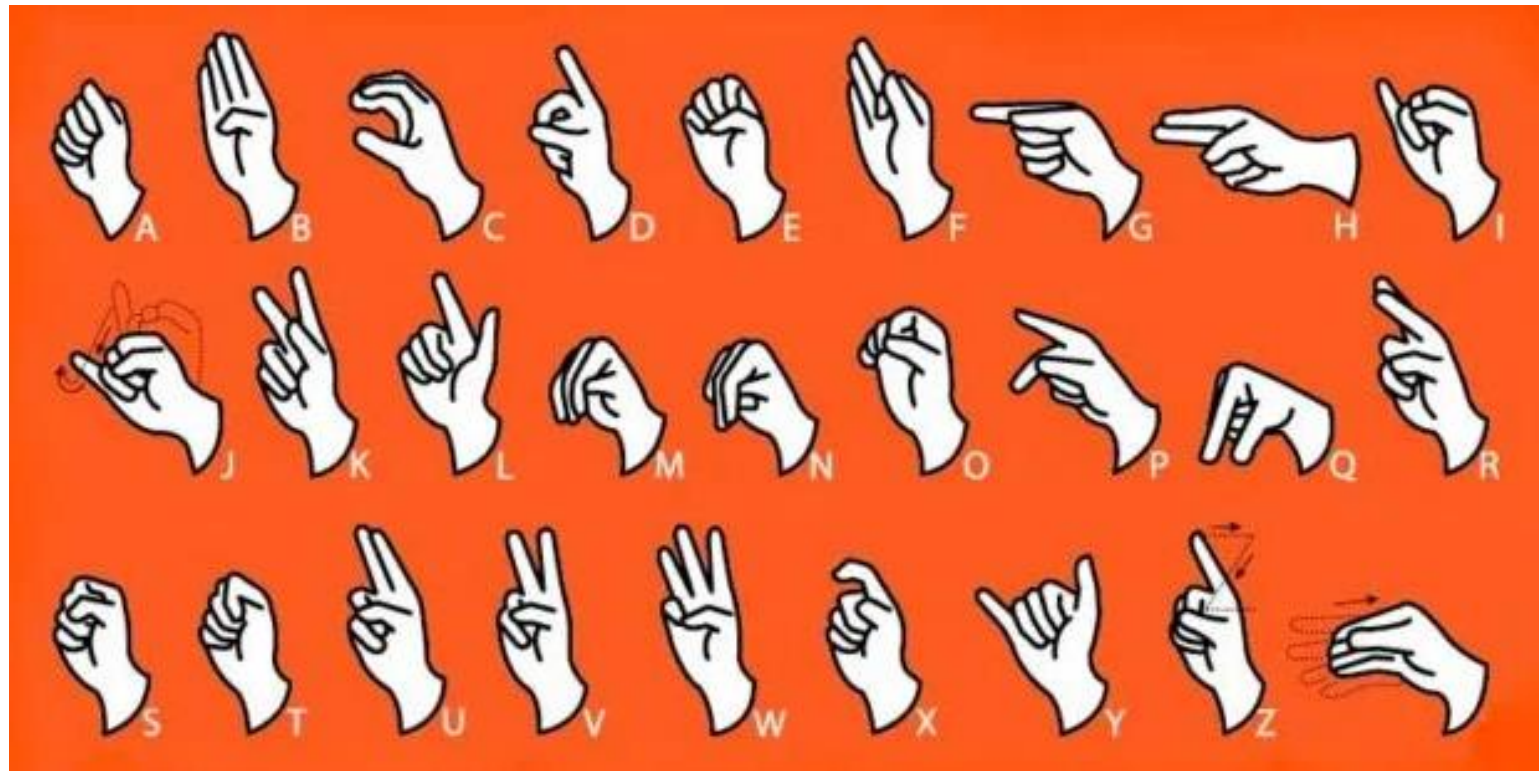
Introduction to American Sign Language

- **Sign Language** is a visual mode of communication.
- Sign language is the **third most used language** in the world!



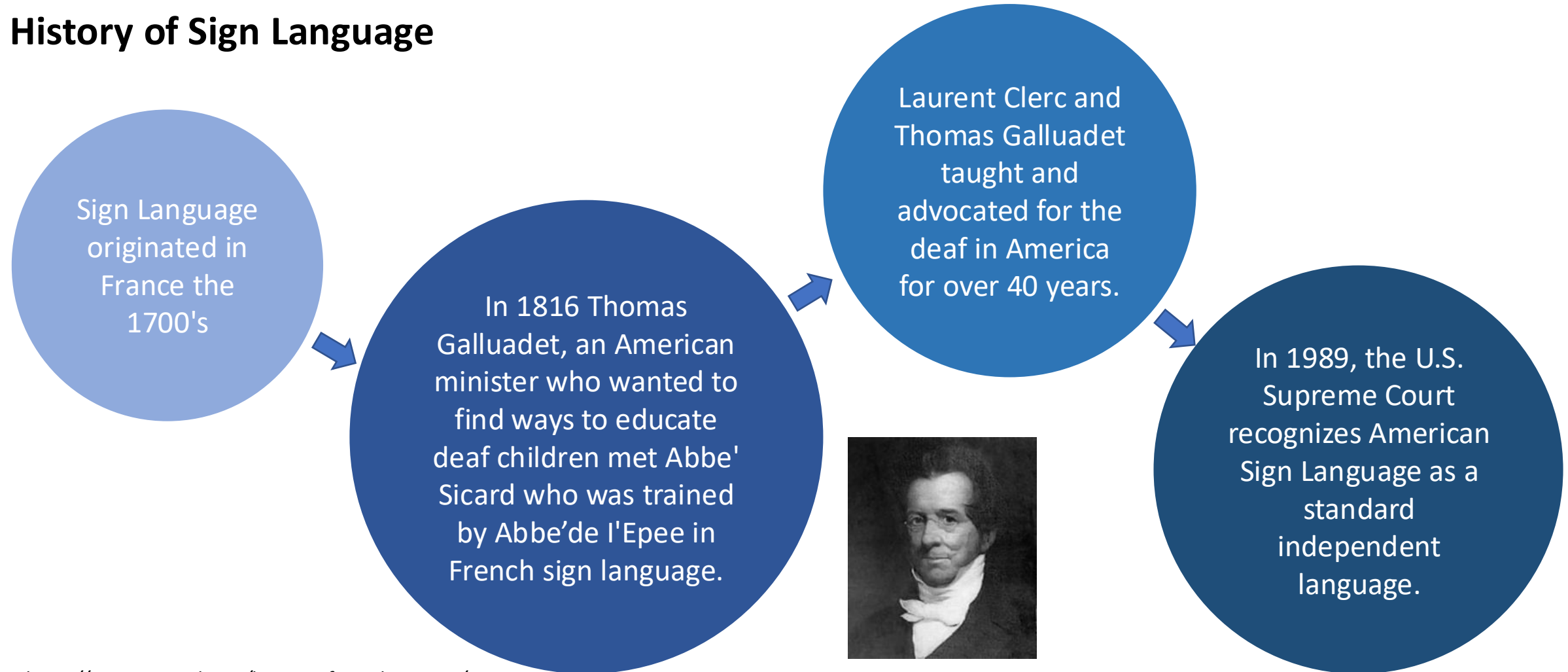
Introduction to American Sign Language

- **American Sign Language (ASL)** is a natural language that serves as the predominant sign language of Deaf communities in the United States of America.
- ASL is expressed by movements of the hands and face.



Introduction to American Sign Language

History of Sign Language



Source: <https://www.startasl.com/history-of-sign-language/>



Introduction to American Sign Language



Source: <https://learnhowtosign.org>



Impact of ASL

Over 70 million deaf people use sign languages as their preferred communication form.

The number of deaf and hard-of-hearing individuals in the United States is about 3.6% of the population (~11.8 million).

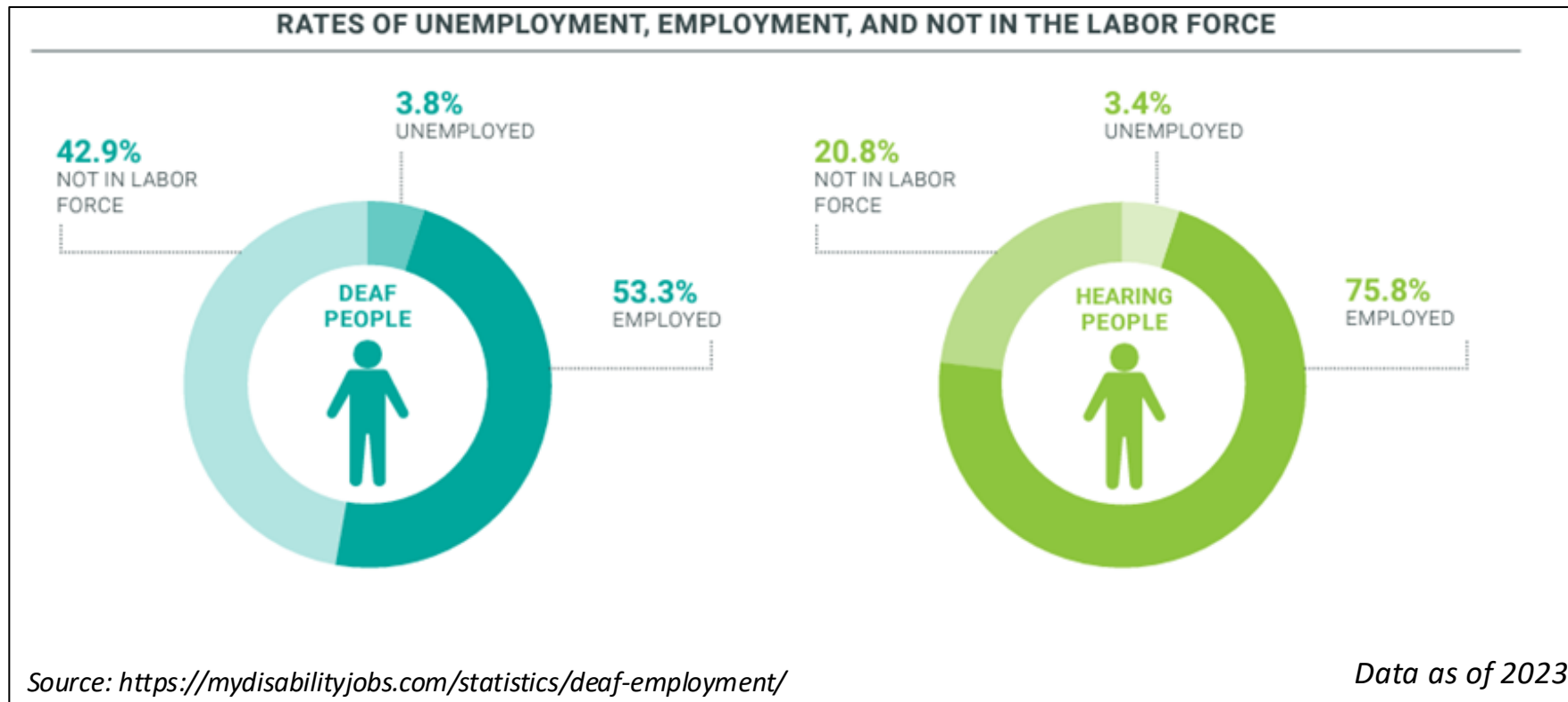
There are only 2,300 sign language interpreters in the USA.

There are 300 different sign languages and 130 that are recognized worldwide.

Source: <https://www.gracesigns.org/>

Impact of ASL on the Society

- For the deaf community, sign language equals rights.
- Engaging and learning sign language is crucial to being inclusive.
- ASL has enabled increased employment rates in people with a hearing disability.



ASL Interpreter Apps



Top Universities for Deaf Students



World leader in Deaf Education

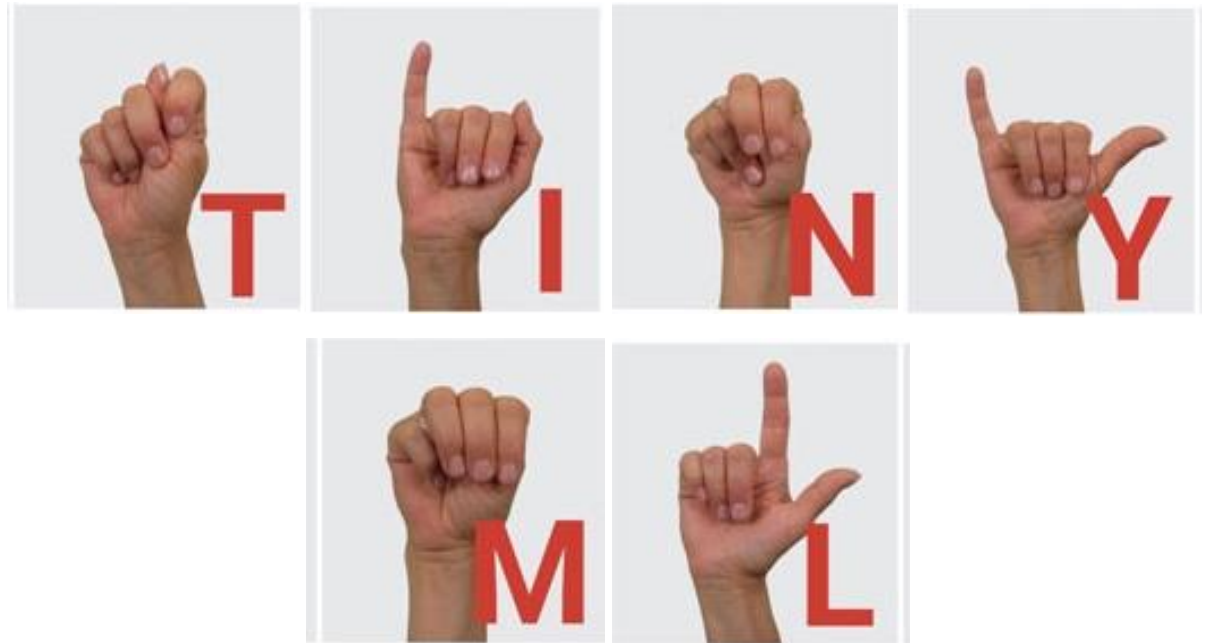
RIT | National Technical
Institute for the Deaf



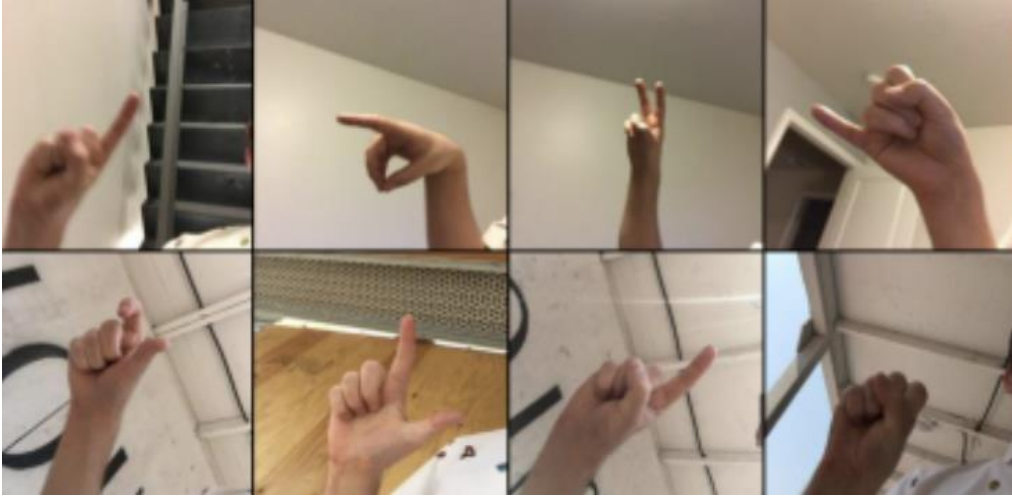
Source: <https://www.healthyhearing.com/report/52682-Top-universities-for-deaf-students>

ASL and TinyML

- How can we use machine learning for ASL?
- Why do we need TinyML for ASL recognition? What are the advantages that TinyML provides over other ASL recognition models?
- What can be possible input data and processing techniques?



ASL Recognition Techniques



Vision-based Methods

Camera image capture
Motion based Gesture recognition
Template Matching

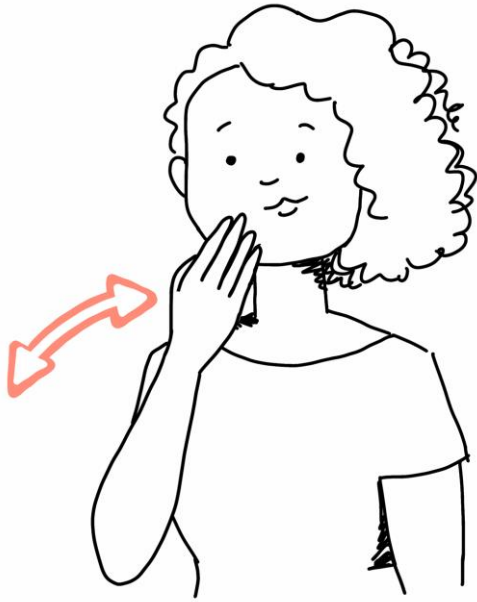


Sensor-based Methods

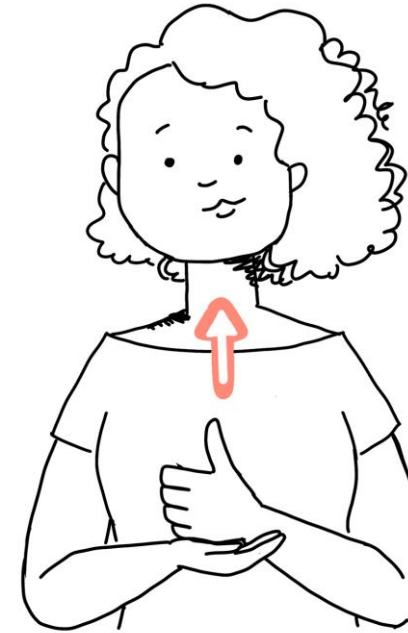
Data Gloves
Accelerometers and gyroscopes
Electromyography (EMG) sensors

Sensing-based Method –Input Data

ASL Gestures

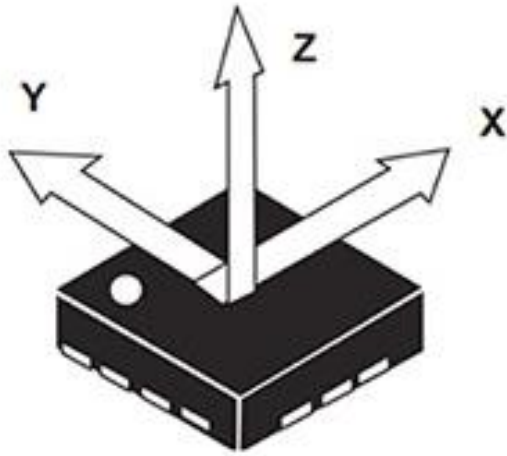


Thank you

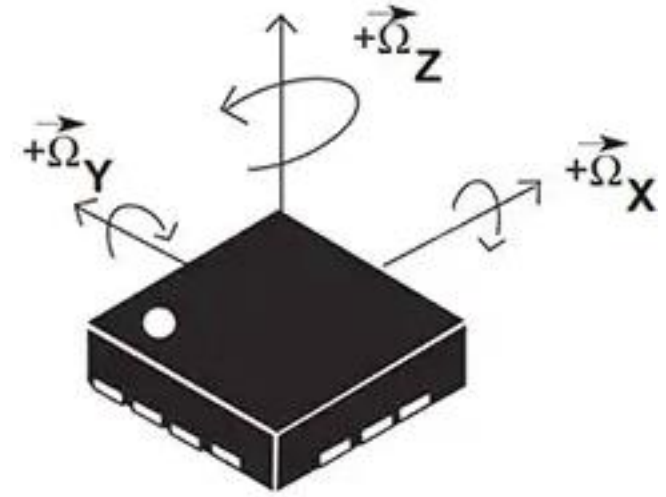


Help

Accelerometer v. Gyroscope



- The accelerometer measures linear acceleration along the x, y, and z axis.
- Senses axis orientation

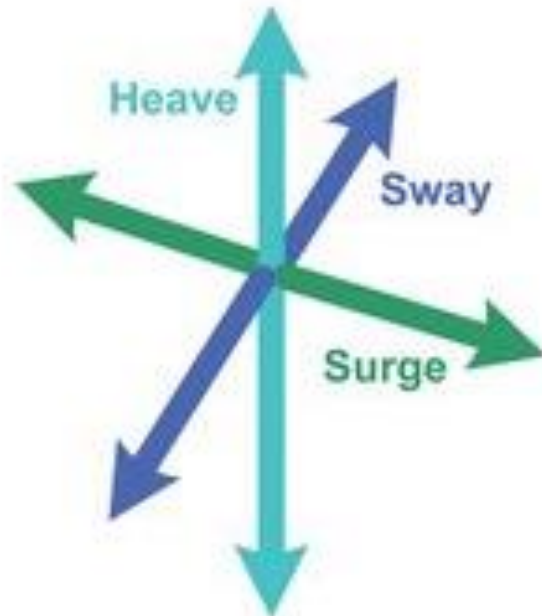


- The gyroscope measures the rate of change of angular velocity over time (angular changes).
- Senses angular orientation

Source: <https://github.com/billiyz/tinyml-on-the-edge>

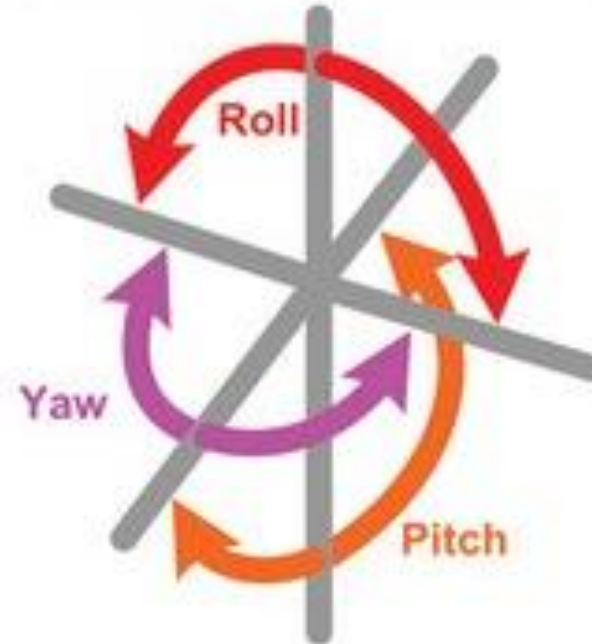
Degrees of Freedom

Translational Movement
in Three Perpendicular Axes



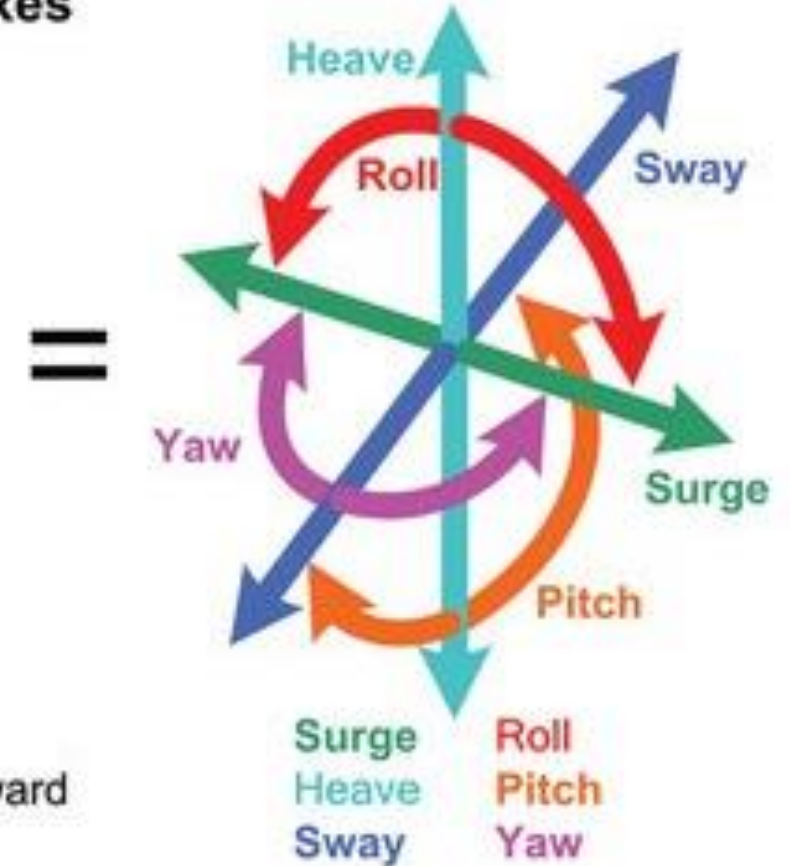
Surge: Moving forward/backward
Heave: Moving up/down
Sway: Moving left/right

Rotational Movement
about Three Perpendicular Axes



Roll: Tilting side to side
Pitch: Tilting forward and backward
Yaw: Turning left and right

Six Degrees of Freedom

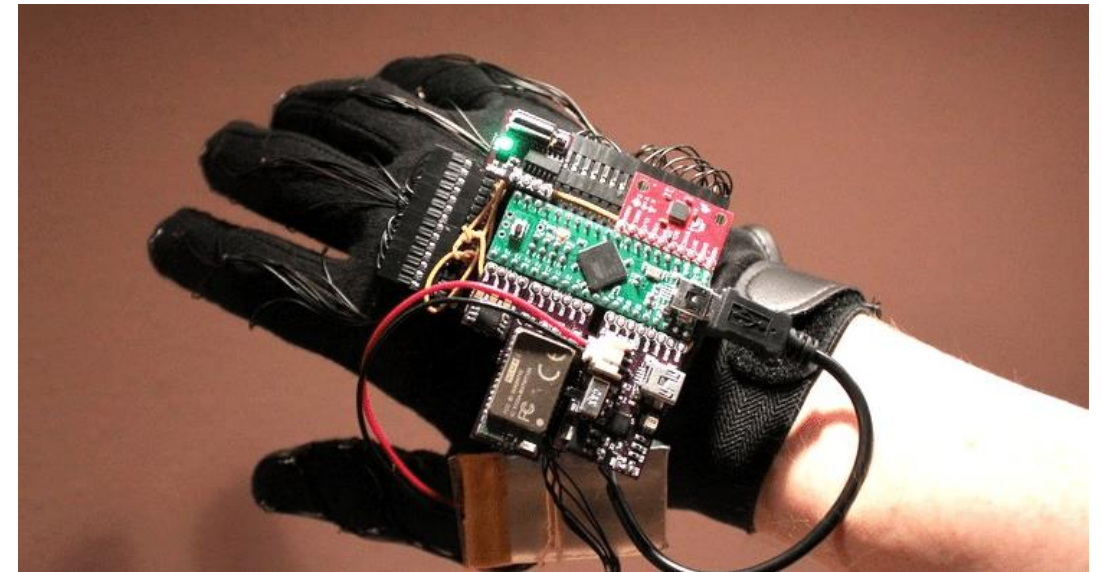
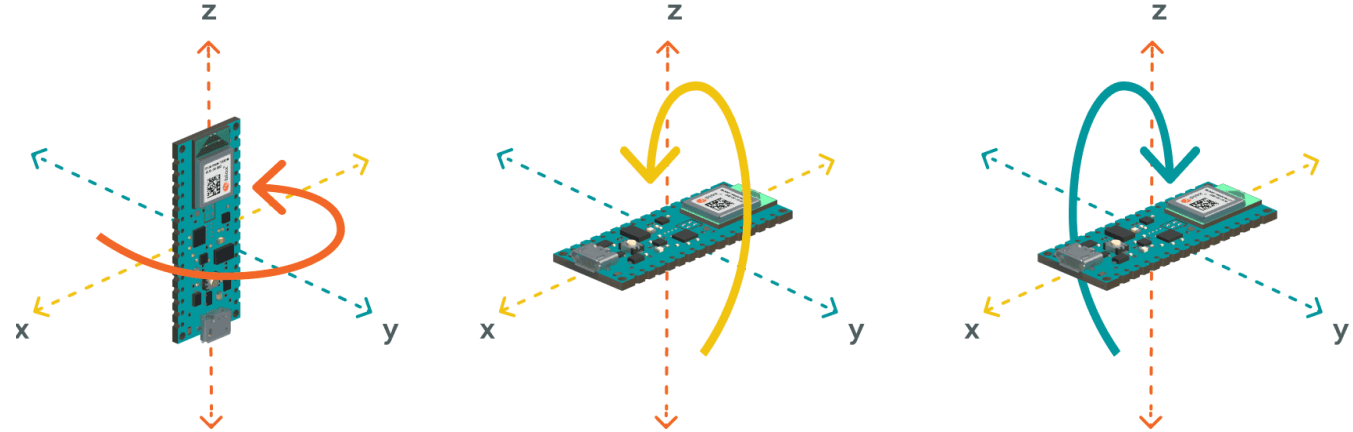
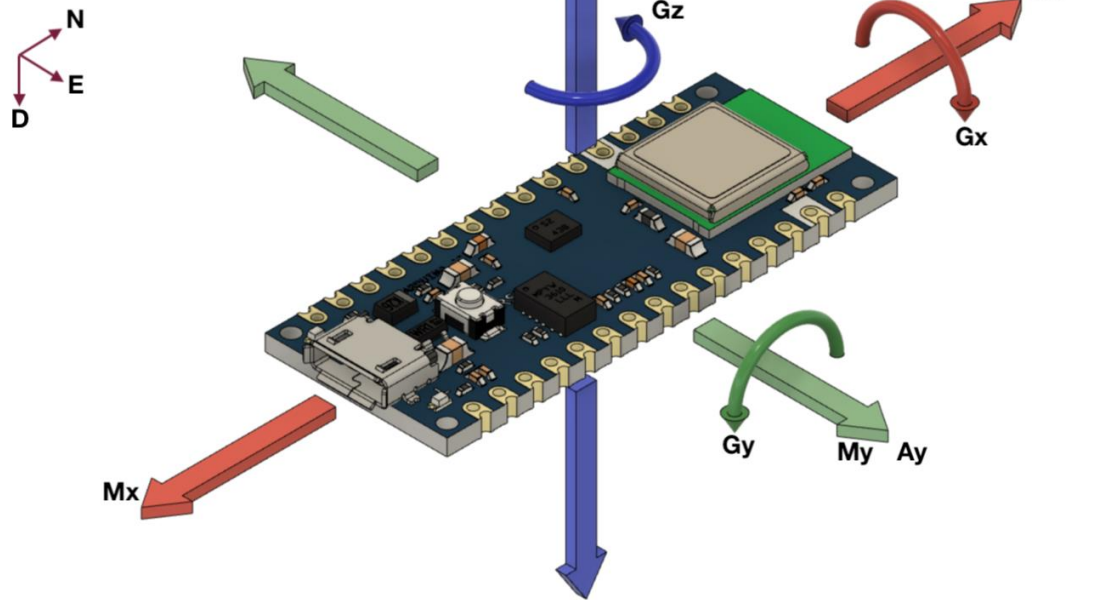


Surge
Heave
Sway

Roll
Pitch
Yaw

Gyroscope on Arduino Nano 33 BLE

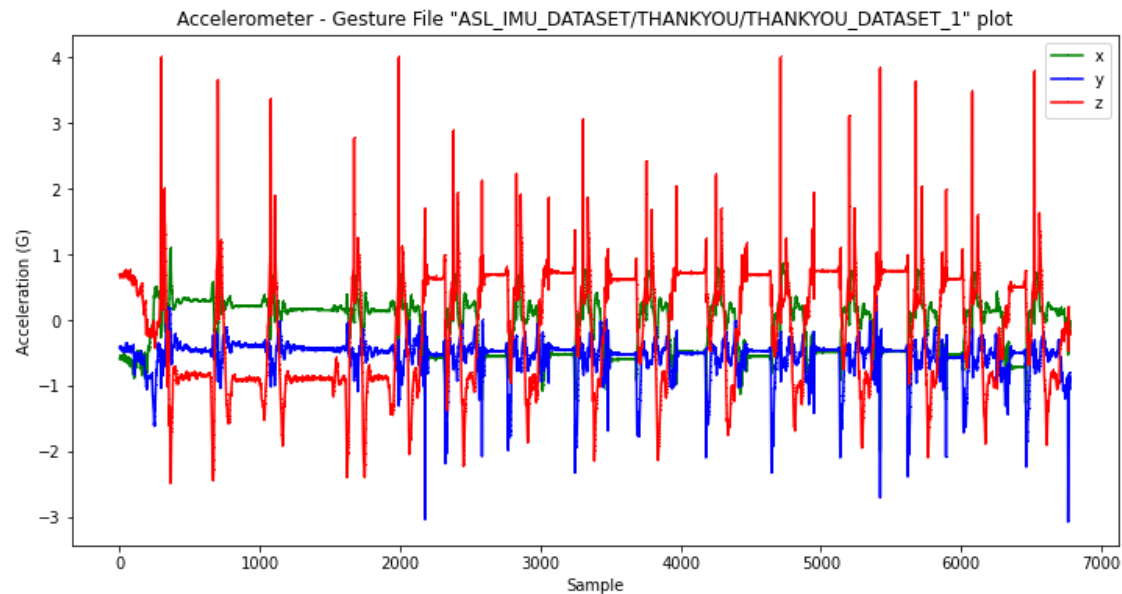
$a_N = a_x, a_E = a_y, a_D = -a_z$
 $g_N = g_x, g_E = g_y, g_D = -g_z$
 $m_N = -m_x, m_E = m_y, m_D = -m_z$



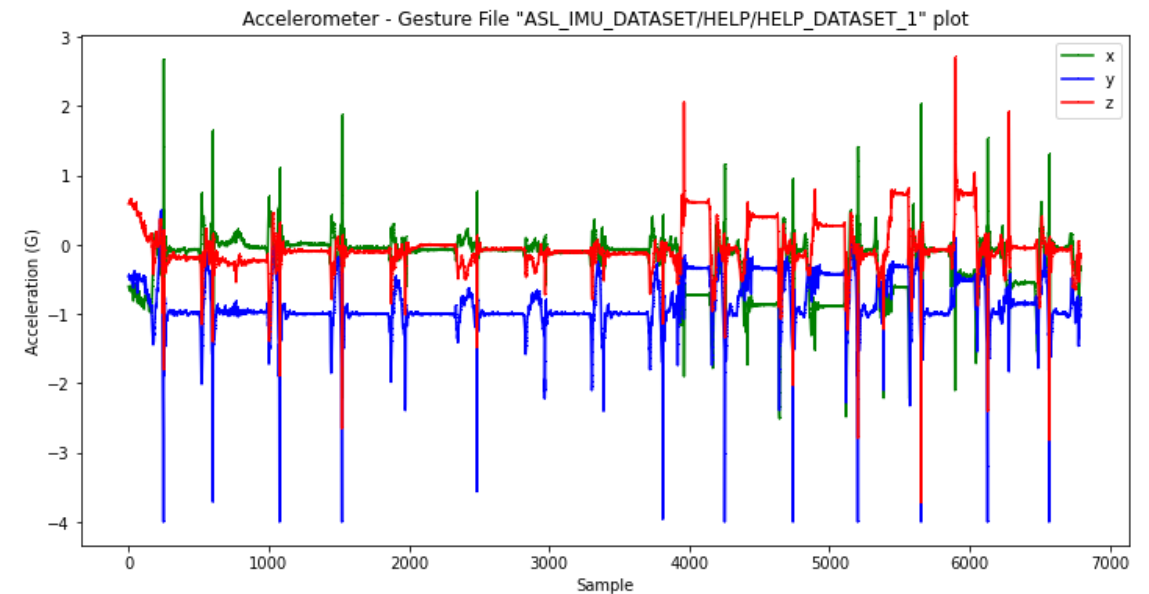
Source: <https://docs.arduino.cc/tutorials/nano-33-ble/imu-gyroscope>

Sensing-based Method - Dataset and Features

ASL IMU Dataset - Accelerometer



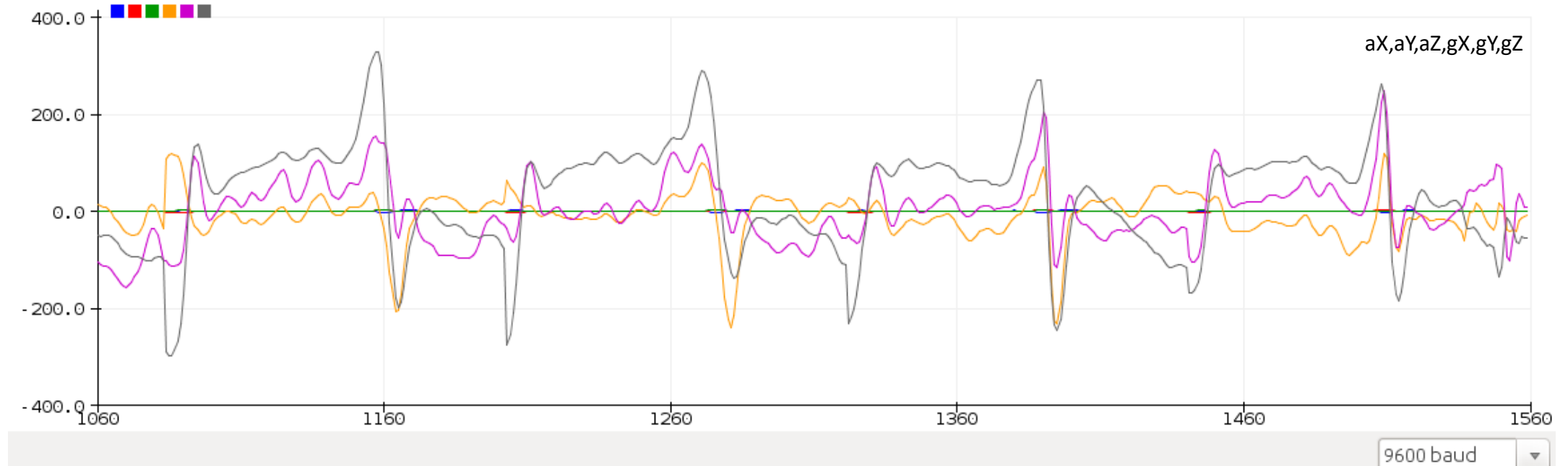
Thank You



Help

Sensing-based Method - Dataset and Features

ASL IMU Dataset – Accelerometer and Gyroscope



Thank You

Vision-based Method – Dataset and Features

Sign Language MNIST

- No cases for the letters J & Z (Reason: J & Z require motion)
- Grayscale Images
- Pixel Values ranging from 0 to 255
- Each image contains 784 Pixels



Data Pre-Processing

- ASL model must be able to work with real time data considerations:
 - Human hand need not always be straight while using gestures
 - There will be background images and disturbances



Dataset with and without a clear background

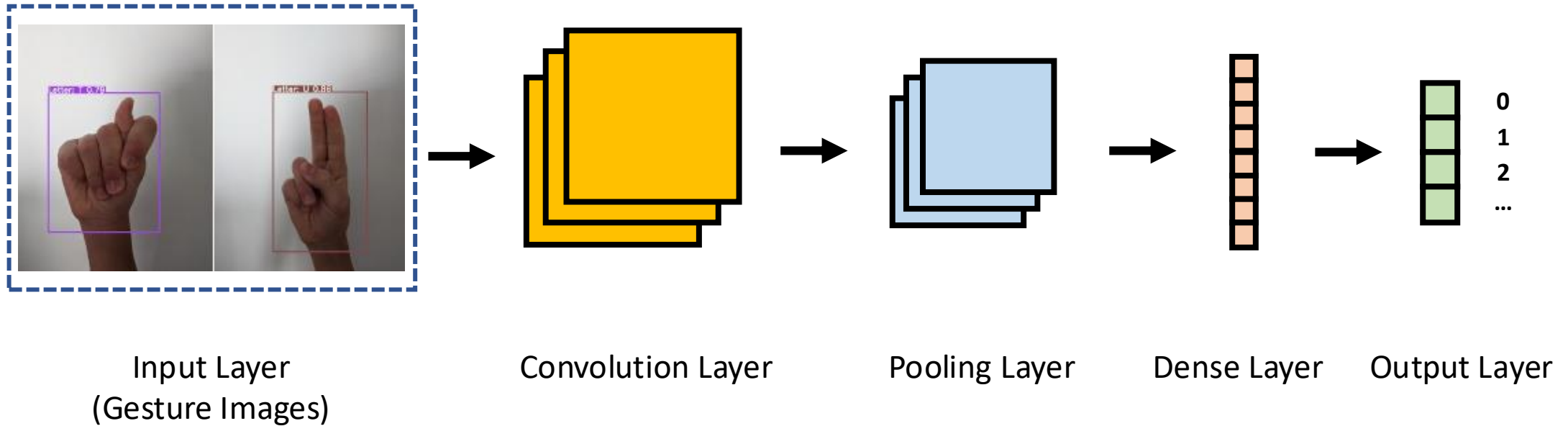


Data Augmentation to avoid overfitting



CNN for ASL

Convolutional Neural Networks (CNNs)

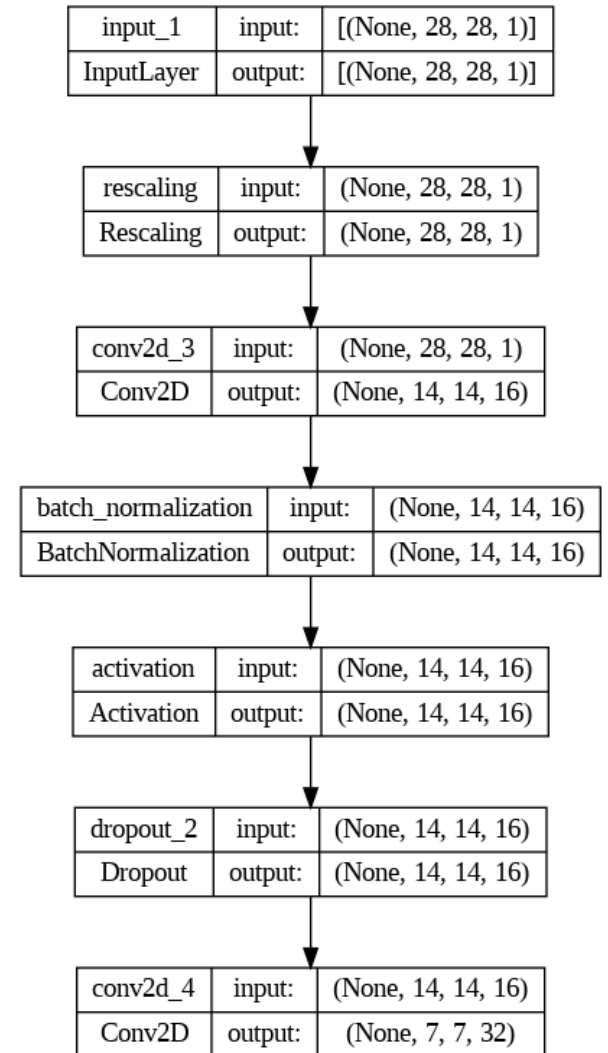


Example CNN Model for Image-based ASL Recognition

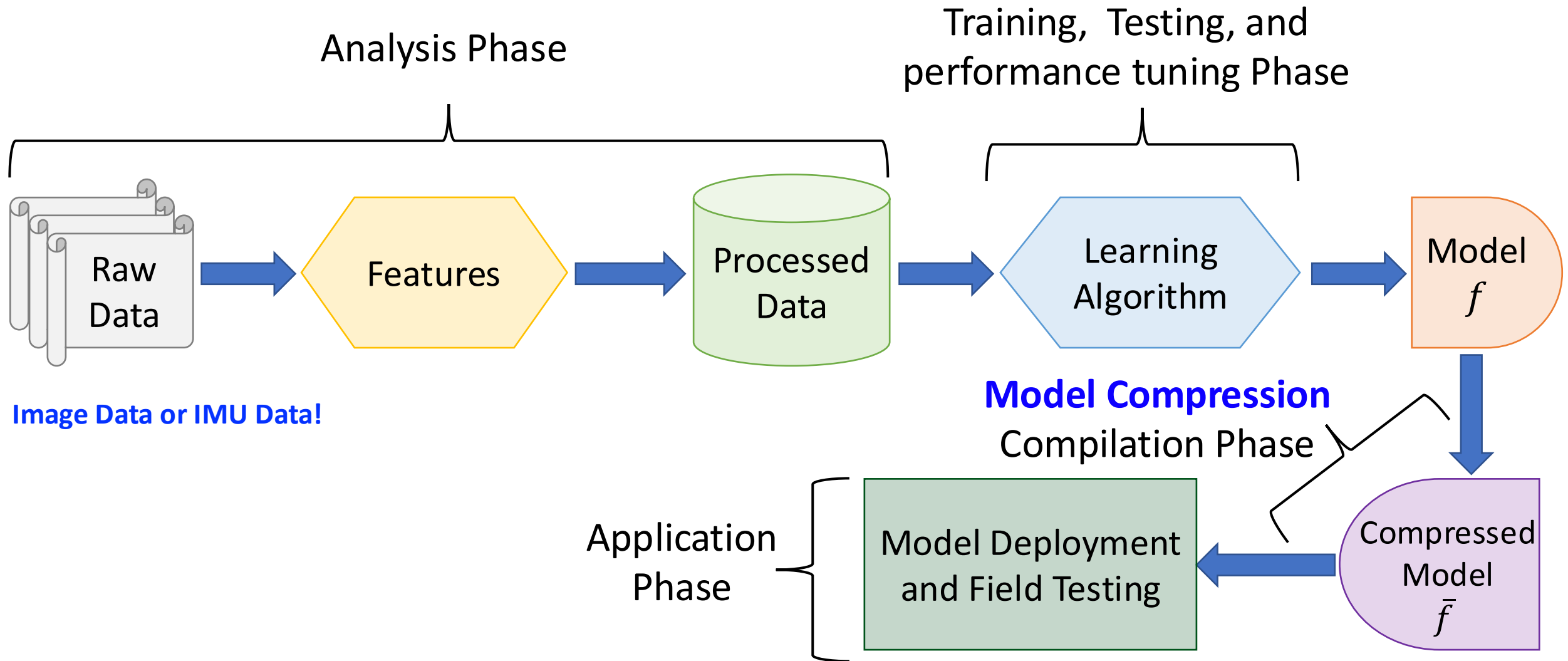
- An example CNN model for classifying ASL alphabets
- Input image size: **28 x 28**
- Output classes: **26**

Note:

- The *Arduino Nano 33 BLE* has a *camera module* which takes *input images in 32 x 32*.
- Necessary *data pre-processing* must be done before training your *CNN models* for image classification/object detection tasks.



A Schematic View of TinyML Pipeline



Today's Lab

- Both Software and Hardware Lab!
- Training models to recognize the alphabet
- Quantize the model for TinyML
- Deploy the model to the Arduino board

